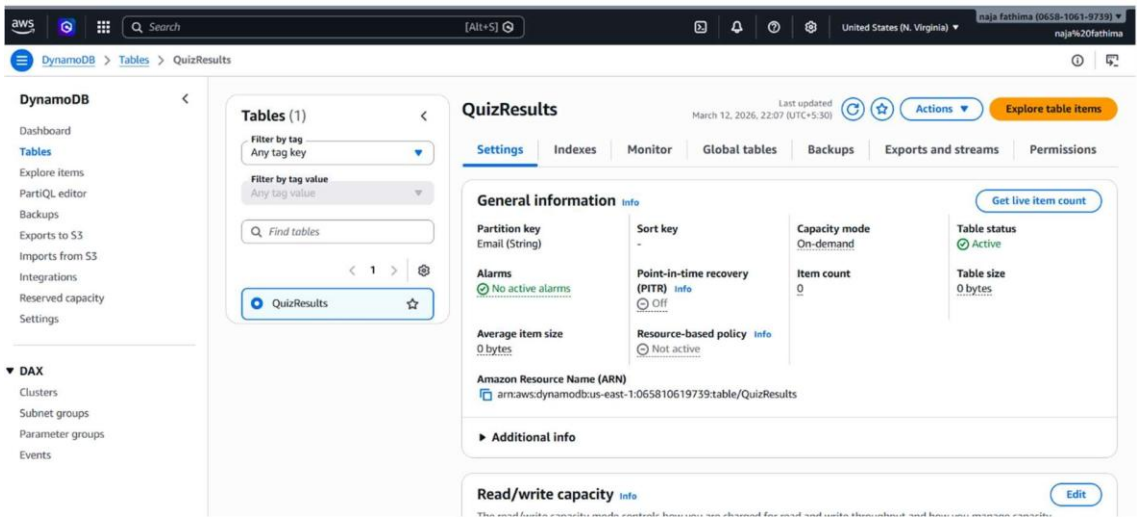
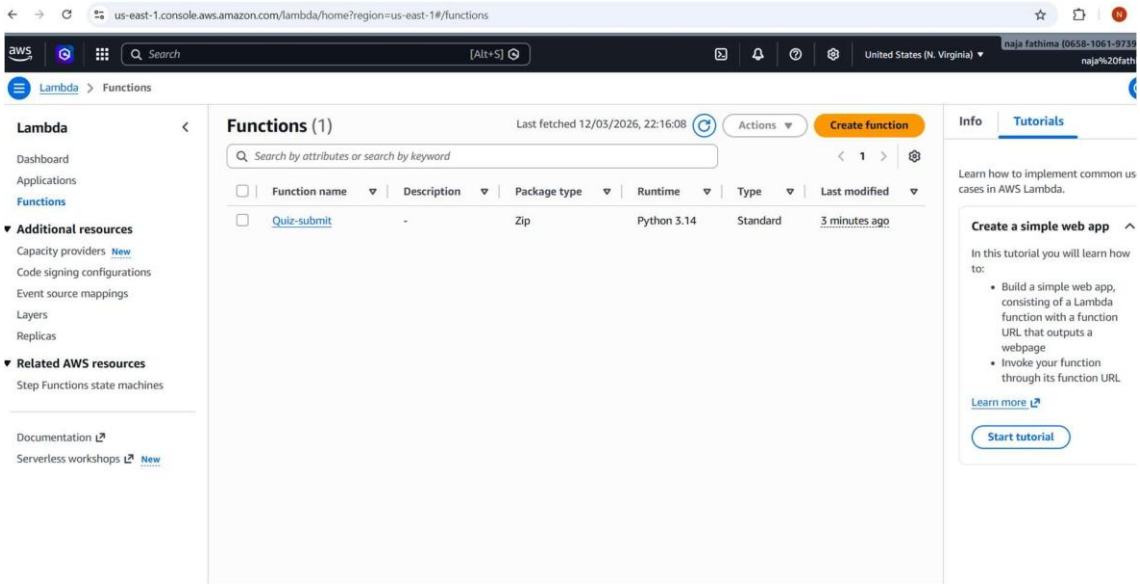
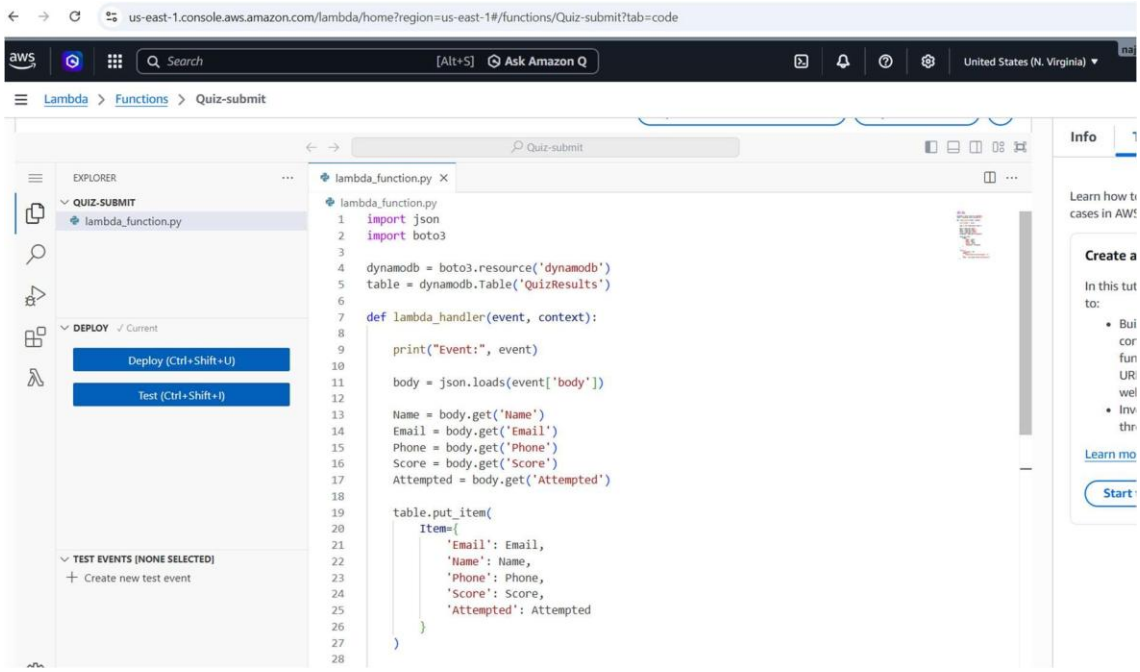
Step 1: Created DynamoDB Table (Database)

Added Email as partition Key

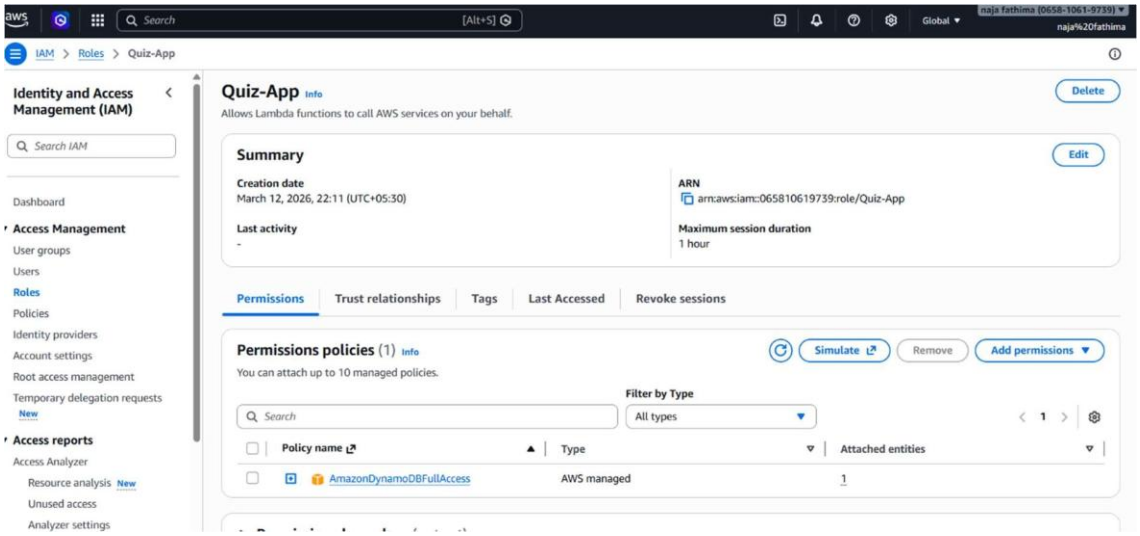


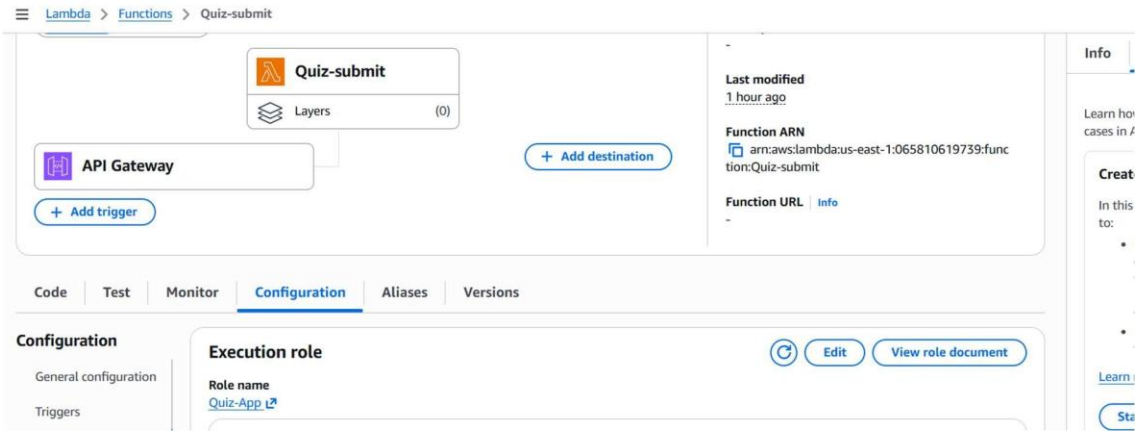
Step 2: Created Lambda Function (Backend)



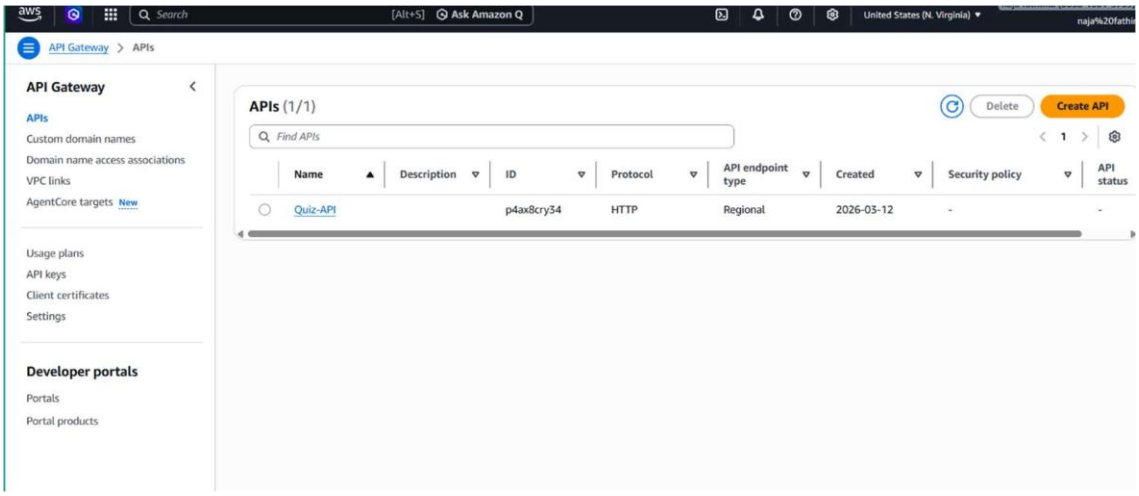


Step 3: Created and attached IAM role for Lambda to access DynamoDB

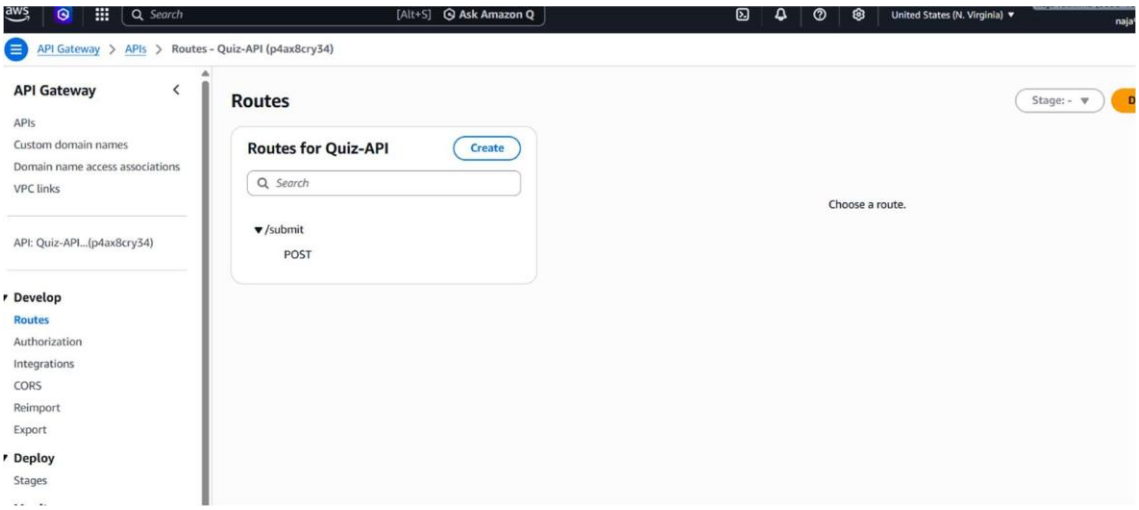




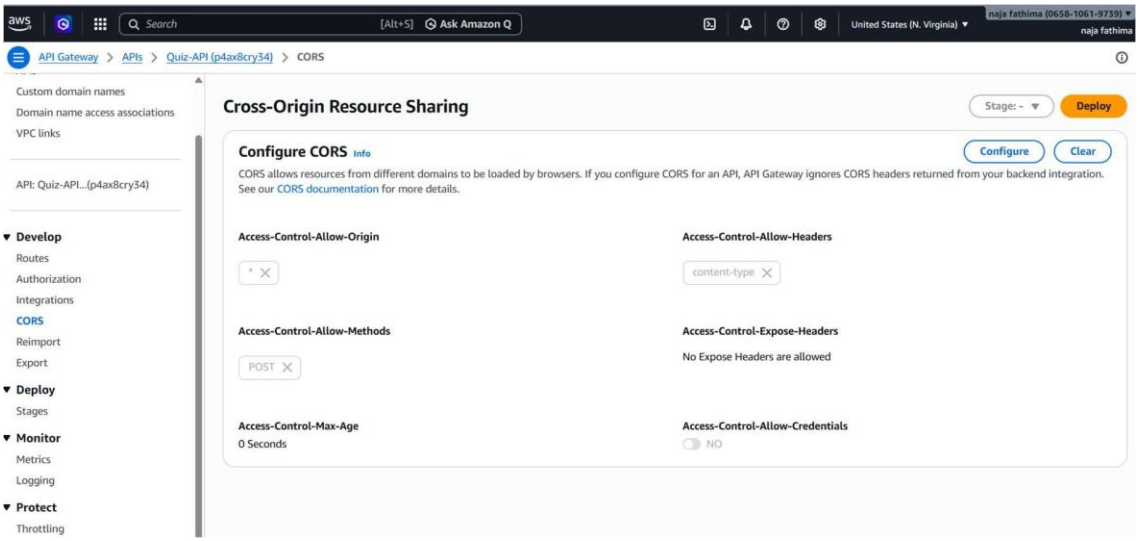
Step 4:Created API Gateway



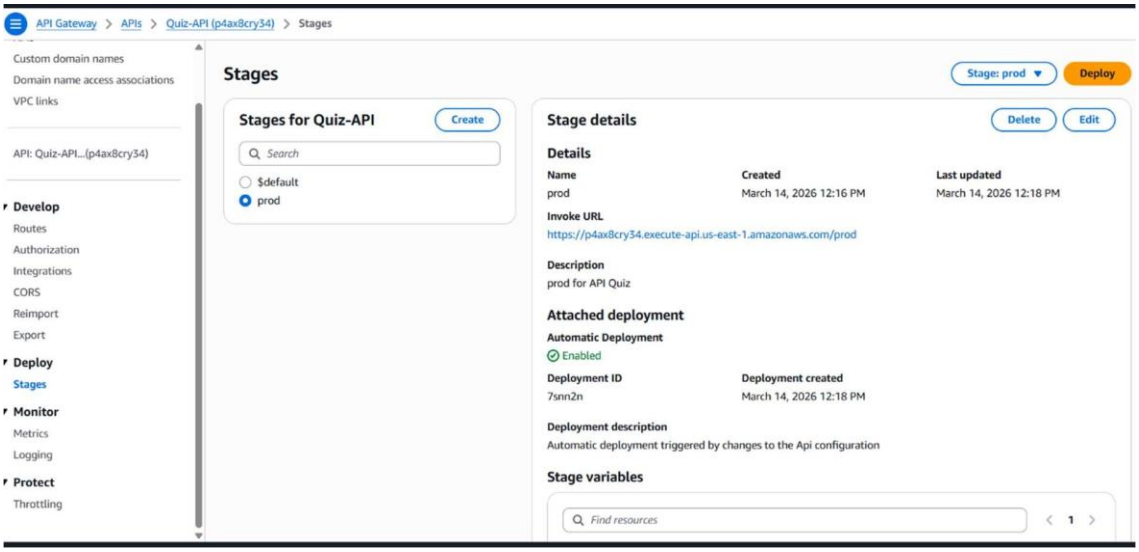
Added Routes



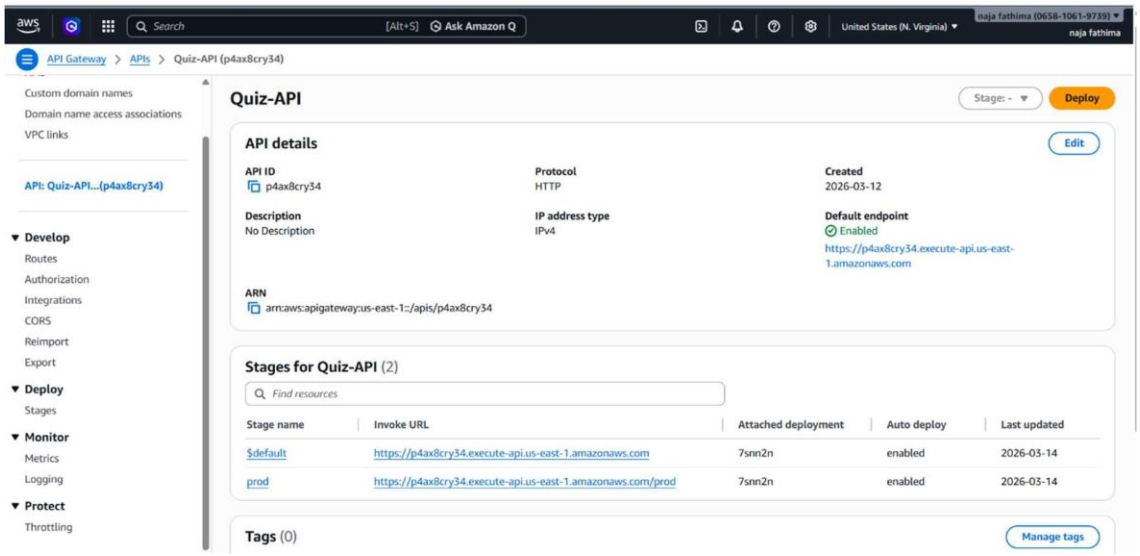
Enable CORS



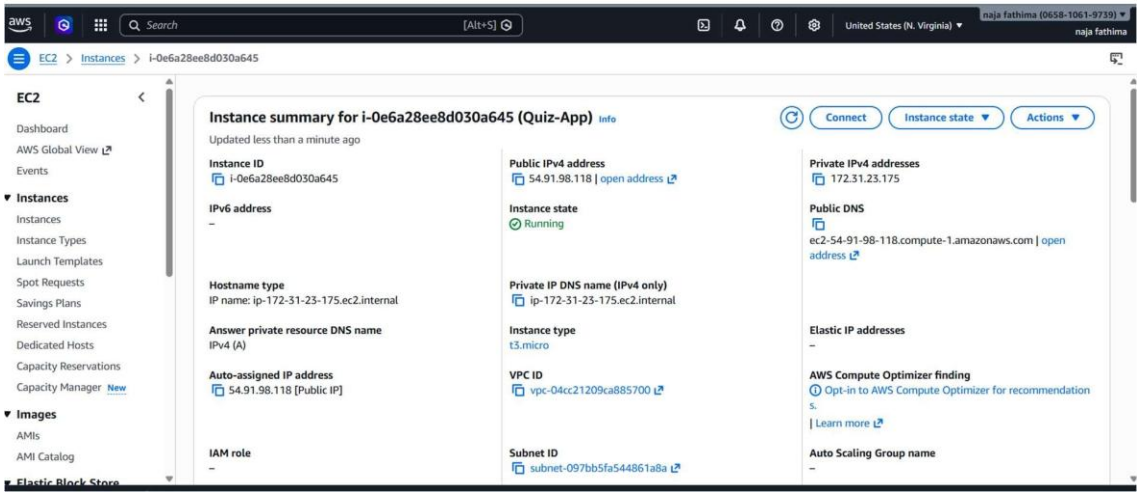
Configured Stages



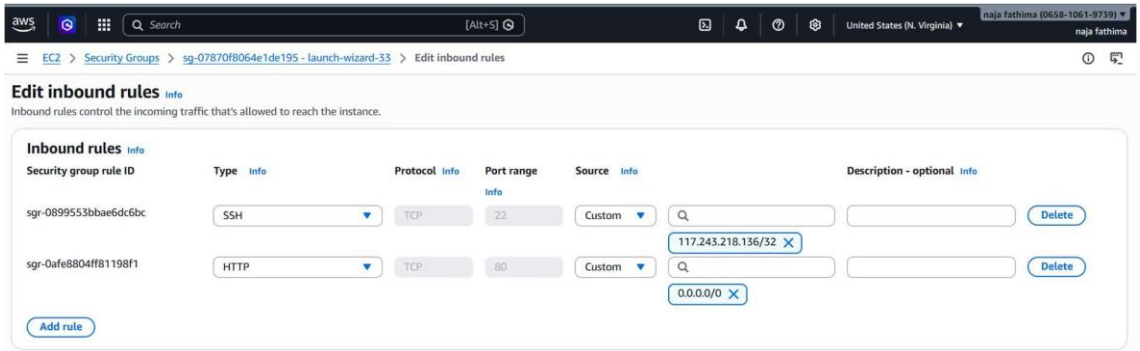
API URL



Step 5:Created an EC2 Instance (Frontend)



Allowed HTTP(80) and SSH restricted to admin IP



Installed Nginx on EC2

```
      _/m/'
[ec2-user@ip-172-31-23-175 ~]$ sudo dnf install nginx -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
================================================================================================================================
 Package                                        Architecture                          Version
================================================================================================================================
Installing:
 nginx                                          x86_64                                1:1.28.2-1.amzn2023.0.1
Installing dependencies:
 generic-logos-httpd                            noarch                                18.0.0-12.amzn2023.0.3
 gperftools-libs                                x86_64                                2.9.1-1.amzn2023.0.3
 libunwind                                      x86_64                                1.4.0-5.amzn2023.0.3
 nginx-core                                     x86_64                                1:1.28.2-1.amzn2023.0.1
 nginx-filesystem                               noarch                                1:1.28.2-1.amzn2023.0.1
 nginx-mimetypes                                noarch                                2.1.49-3.amzn2023.0.3

Transaction Summary
================================================================================================================================
Install  7 Packages

Total download size: 1.1 M
Installed size: 3.7 M
Downloading Packages:
(1/7): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm
(2/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm
(3/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm
(4/7): nginx-1.28.2-1.amzn2023.0.1.x86_64.rpm
(5/7): nginx-core-1.28.2-1.amzn2023.0.1.x86_64.rpm
(6/7): nginx-filesystem-1.28.2-1.amzn2023.0.1.noarch.rpm
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm
--------------------------------------------------------------------------------------------------------------------------------
Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :
  Running scriptlet: nginx-filesystem-1:1.28.2-1.amzn2023.0.1.noarch
  Installing       : nginx-filesystem-1:1.28.2-1.amzn2023.0.1.noarch
  Installing       : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch
  Installing       : libunwind-1.4.0-5.amzn2023.0.3.x86_64
  Installing       : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64
  Installing       : nginx-core-1:1.28.2-1.amzn2023.0.1.x86_64
  Installing       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
  Installing       : nginx-1:1.28.2-1.amzn2023.0.1.x86_64
```

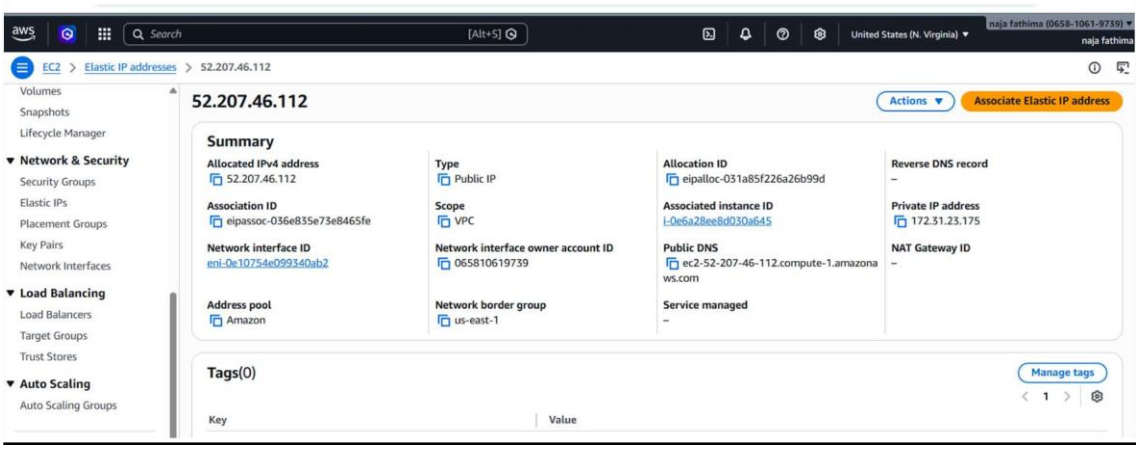
```
  nginx-core-1:1.28.2-1.amzn2023.0.1.x86_64                    nginx-filesystem-1:1.28.2-1.amzn2023.0.1.noa

Complete!
[ec2-user@ip-172-31-23-175 ~]$ sudo systemctl enable --now nginx
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/ngi
[ec2-user@ip-172-31-23-175 ~]$
```

Uploaded Quiz Frontend

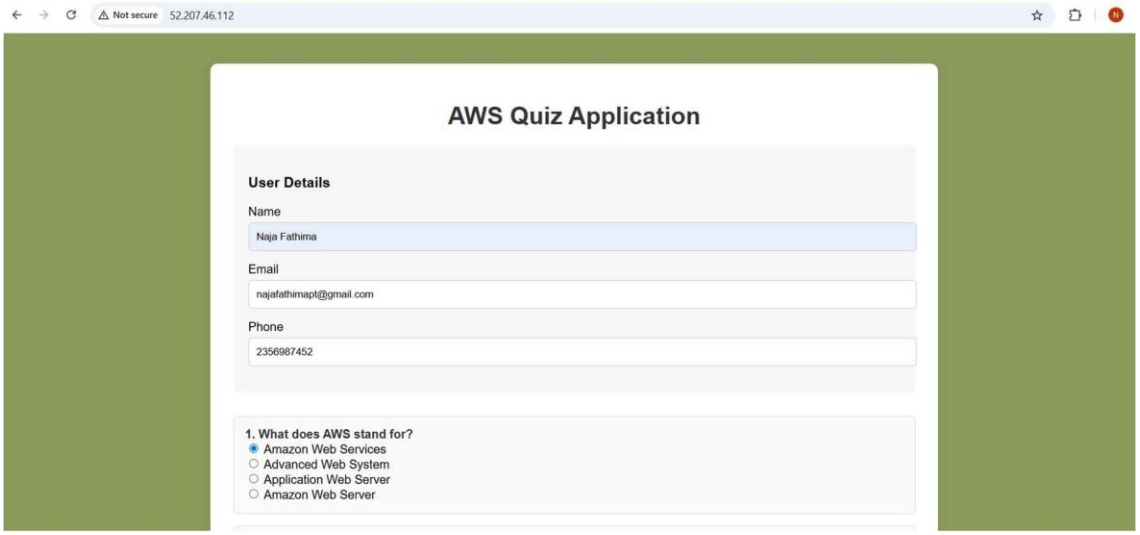
```
[ec2-user@ip-172-31-23-175 ~]$ sudo vi /usr/share/nginx/html/index.html
[ec2-user@ip-172-31-23-175 ~]$ sudo systemctl restart nginx
[ec2-user@ip-172-31-23-175 ~]$ |
```

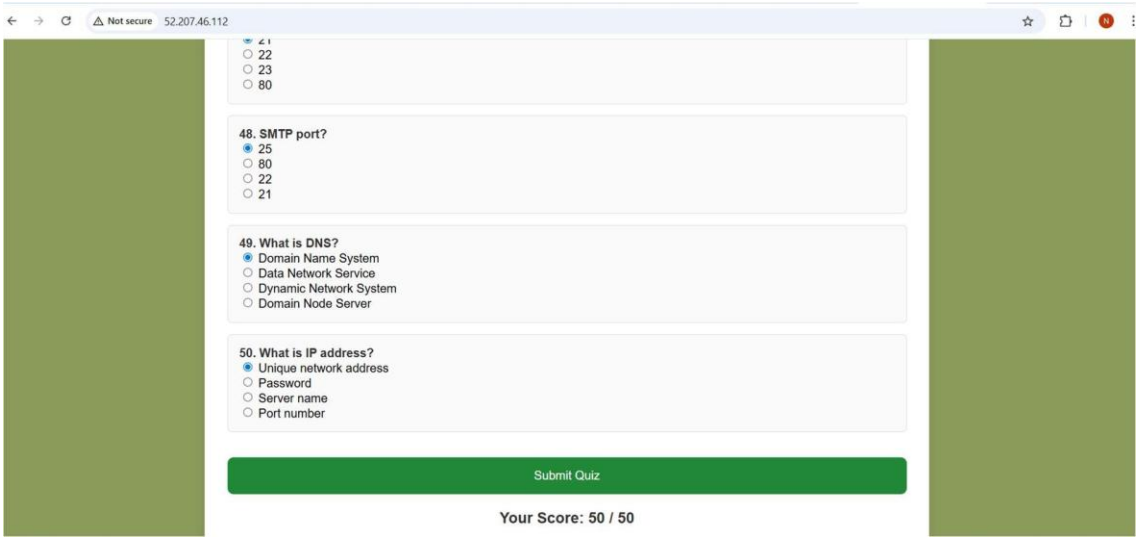
```
ec2-user@ip-172-31-23-175:~
{q:"47. FTP port?",a:["21","22","23","80"],c:0},
{q:"48. SMTP port?",a:["25","80","22","21"],c:0},
{q:"49. What is DNS?",a:["Domain Name System","Data Network Service","Dynamic Network System","Domain Node Server"],c:0},
{q:"50. What is IP address?",a:["Unique network address","Password","Server name","Port number"],c:0}

];

const quizDiv = document.getElementById("quiz");

questions.forEach((item,index)=>{
let html = `<div class="question"><b>${item.q}</b><br>`;
item.a.forEach((opt,i)=>{
html += `<input type="radio" name="q${index}" value="${i}"> ${opt}<br>`;
});
html += "</div>";
quizDiv.innerHTML += html;
});

function submitQuiz(){

let Score = 0;

questions.forEach((item,index)=>{
let ans = document.querySelector(`input[name="q${index}"]:checked`);
if(ans && ans.value == item.c){
Score++;
}
});

let Name = document.getElementById("Name").value;
let Email = document.getElementById("Email").value;
let Phone = document.getElementById("Phone").value;

document.getElementById("result").innerHTML =
"Your Score: "+Score+" / "+questions.length;


fetch("https://p4ax8cry34.execute-api.us-east-1.amazonaws.com/prod/submit",{
method:"POST",
headers:{
"Content-Type":"application/json"
},
body:JSON.stringify({
Name:Name,
Email:Email,
Phone:Phone,
Score:Score,
Attempted:50
})
});

}

</script>
```

Step 6: Created Elastic IP for EC2

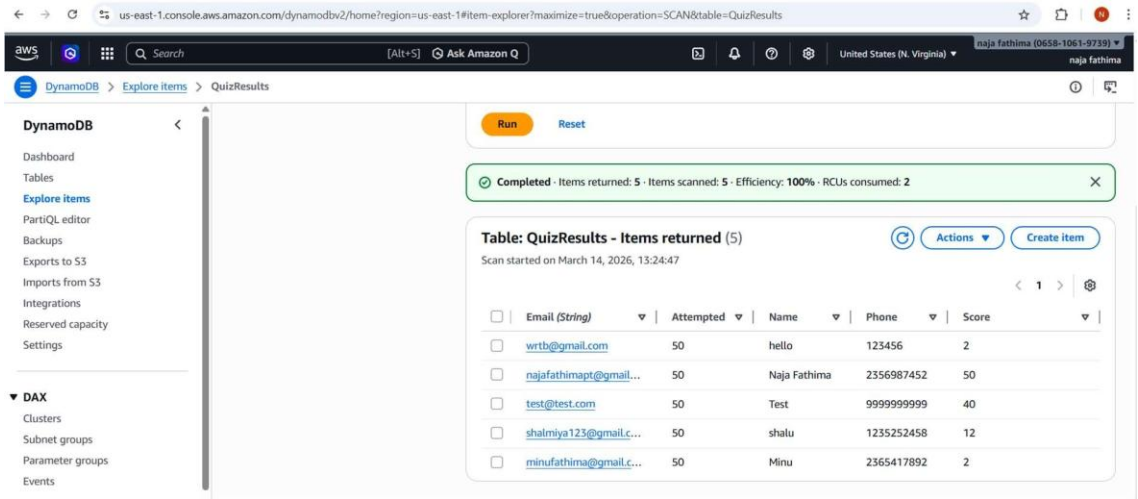


Step 7: Tested the Quiz Applica on using EC2 Elastic public IP





Step 8: Verified Data on DynamoDB


us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#item-explorer?maximize=true&operation=SCAN&table=QuizResults
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
naja fathima (0658-1061-9739)
naja fathima
DynamoDB > Explore items > QuizResults
DynamoDB
Dashboard
Tables
Explore items
PartiQL editor
Backups
Exports to S3
Imports from S3
Integrations
Reserved capacity
Settings
DAX
Clusters
Subnet groups
Parameter groups
Events
Run
Reset
Completed · Items returned: 5 · Items scanned: 5 · Efficiency: 100% · RCUs consumed: 2
Table: QuizResults - Items returned (5)
Scan started on March 14, 2026, 13:24:47
Actions
Create item
Email (String)
Attempted
Name
Phone
Score
wrtb@gmail.com 50 hello 123456 2
najafathimapt@gmail... 50 Naja Fathima 2356987452 50
test@test.com 50 Test 9999999999 40
shalmiya123@gmail.c... 50 shalu 1235252458 12
minufathima@gmail.c... 50 Minu 2365417892 2